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Project in Wireless Communication

Lecture 5: Project overview and cyclic redundancy check (CRC) codes

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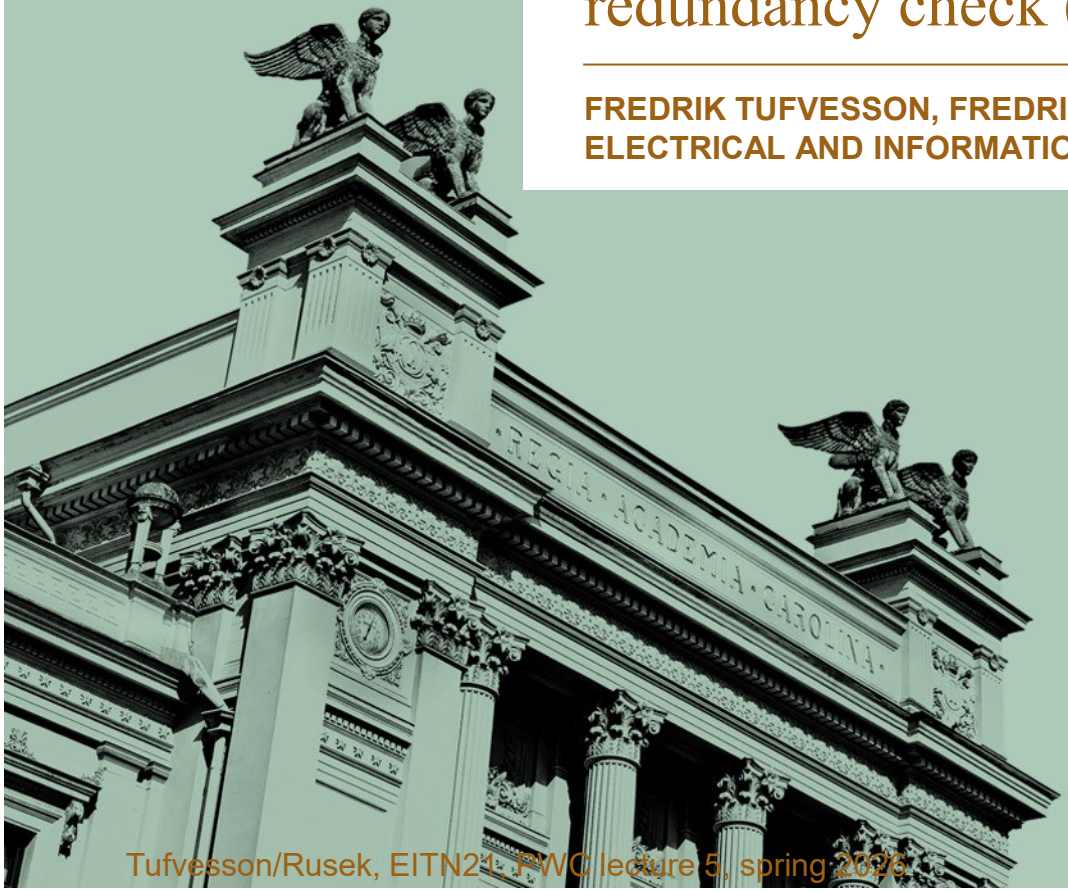
ELECTRICAL AND INFORMATION TECHNOLOGY



Cyclic redundancy check, CRC

A CRC is used for ***error detection***, not for error correction.

Example: Single parity check bit

Suppose one wants to transmit the 5 bits

[0 0 1 0 1]

If one receives the bits

[0 0 **0** 0 1]

This error will pass by un-detected.

CRC Example: Single parity check bit

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This error will pass by un-detected.

We can fix this by adding a single parity bit so that the total number of 1s is always even.

We then have

Parity bit

[0 0 1 0 1 **0**]

Total number of 1s = even

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CRC Example: Single parity check bit

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[0 0 1 0 1]

If one receives the bits

[0 0 **0** 0 1]

This error will pass by un-detected.

We can fix this by adding a single parity bit so that the total number of 1s is always even.

Parity bit

We then have [0 0 1 0 1 **0**]

Total number of 1s = odd

If we receive [0 0 0 0 1 **0**] ***-> not correct***

we know that there has been 1 bit error on the channel, and we will ask for a re-transmission

CRC

The previous example was just meant as illustration, and in reality, much more advanced systems are used. **But, they are based upon the same principle!**

Suppose that we should send *K* bits, [$u_0 \ldots u_{K-1}$]. We denote these by the *D*-transform

$$\mathrm{u}(D) \stackrel{\text{def}}{=} u_{K-1}D^{K-1} + u_{K-2}D^{K-2} + \cdots + u_0$$

Hence, $1 + D + D^6 = [1\ 1\ 0\ 0\ 0\ 0\ 1]$ etc etc

The powers of the indeterminate *D* can be thought of as keeping track of which bit is which. The CRC is represented by another polynomial,

$$\mathrm{c}(D) \stackrel{\text{def}}{=} c_{L-1}D^{L-1} + c_{L-2}D^{L-2} + \cdots + c_0$$

The entire frame of data and CRC is then $x(D) = u(D)D^L + c(D)$, that is

$$x(D) \stackrel{\text{def}}{=} u_{K-1}D^{L+K-1} + u_0D^L + c_{L-1}D^{L-1} \ldots + c_0$$

How to find *c(D)*?

The check bits *c(D)* depend of course on the information bits *u(D)*.

Question: How to find the check bits *c(D)* given a particular set of information bits *u(D)* in a structured fashion?

How to find *c(D)*?

Define a generator polynomial *g(D)* of degree *L*

$$g(D) \stackrel{\text{def}}{=} D^L + g_{L-1}D^{L-1} + \cdots + g_1 D + 1$$

For a given generator polynomial, *g(D)*, the mapping from the information bits, *u(D)*, to the CRC, *c(D)*, is given by

$$\boxed{c(D) = Remainder\left[\frac{u(D)D^L}{g(D)}\right]}$$

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Long Division

$$c(D) = Remainder\left[\frac{u(D)D^L}{g(D)}\right]$$

- This is just an ordinary long division of one polynomial with another.
- All operations are modulo 2. Thus (1+1) mod 2 = 0, and (0-1) mod 2 = 1.
- Subtraction using modulo 2 arithmetic is the same as addition

Example:

$$\underline{D^3 + D^2 + 1}\,\overline{\big)\, D^5 + \qquad\quad D^3}$$

Find remainder of $D^5 + D^3$ *divided with* $D^3 + D^2 + D$

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$$\begin{array}{r|l} & D^2 \\ \hline D^3 + D^2 + 1 & D^5 + \qquad\quad D^3 \end{array}$$

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$$\begin{array}{r|l}
 & D^2 \\
\hline
D^3 + D^2 + 1 & D^5 + \qquad D^3 \\
 & + D^5 + D^4 + \qquad D^2 \\
\hline
 & \qquad D^4 + D^3 + D^2
\end{array}$$

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Example:

$$\begin{array}{r|l}
 & D^2 + D \\
\hline
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 & + D^5 + D^4 + \qquad D^2 \\
\hline
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\hline
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\hline
 & D^2 + D = Remainder
\end{array}
$$

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Example:

quotient

$$
\begin{array}{r|l}
 & D^2 + D \\
\hline
D^3 + D^2 + 1 & D^5 + D^3 \\
 & + D^5 + D^4 + D^2 \\
\hline
 & D^4 + D^3 + D^2 \\
 & + D^4 + D^3 + D \\
\hline
 & D^2 + D = Remainder
\end{array}
$$

$$\boldsymbol{D^5 + D^3 = (D^2 + D)(D^3 + D^2 + 1) + (D^2 + D)}$$

CRC

Let *z(D)* denote the quotient. We then have:

$$u(D)D^L = g(D)z(D) + c(D)$$

Subtract *c(D)* from both sides and use ”+” = ”-” in modulo 2 arithmetic

$$x(D) = u(D)D^L + c(D) = g(D)z(D)$$

Thus, all valid code words *x(D)* are divisible by the generator polynomial *g(D)*

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Receiver operation

Assume *x(D)* is transmitted and that *y(D)* is received. Let the errors on the channel be *e(D)*.

Hence, $y(D)=x(D)+e(D)$.

The receiver knows that a valid *y(D)* should leave no remainder if divided by *g(D)*.

So, the receiver declares:

$$\textbf{ACK if } \boldsymbol{Remainder}\,\frac{y(D)}{g(D)} = 0$$

$$\textbf{NACK if } \boldsymbol{Remainder}\,\frac{y(D)}{g(D)} \neq 0$$

When does it fail?

Since we have shown that *x(D)* is divisible by *g(D)*,

$$Remainder\,\frac{y(D)}{g(D)} = Remainder\left[\frac{x(D)+e(D)}{g(D)}\right] = Remainder\,\frac{e(D)}{g(D)}$$

If no errors occur, i.e., *e(D)*=0, then this remainder is zero, and the receiver declares a successful transmission.

If *e(D)* is not zero, the receiver fails to detect the error only if Rem[*e*(*D*)/*g*(*D*)]=0.

This is the same as saying that *e(D)* is a valid code word, i.e.,

$$e(D) = g(D)z(D)$$

For some non-zero polynomial *z(D)*

When does it fail?

Suppose that a single error occurs, i.e., $e(D) = D^i$, for some integer *i.*

We have an un-detected error if and only if

$$e(D) = g(D)z(D) \quad \text{for some } z(D)$$

But since *g(D)* have at least two non-zero terms (1 and D^L), so must *g(D)z(D)* have.

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But since *g(D)* have at least two non-zero terms (1 and D^L), so must *g(D)z(D)* have.

Hence, *g(D)z(D)* cannot possibly equal D^i and we can conclude

All single event errors are detectable

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Burst errors

We next consider bursts of errors e=[...000 $\underbrace{110\ 10110\ \ldots 01101}_{P}$ 000....]

We know that it will pass un-detected if and only if $e(D) = g(D)\, z(D)$ for some $z(D)$

But,

$$\underbrace{(D^L + \cdots + 1)}_{g(D)}\underbrace{(D^j + \cdots + D^i)}_{z(D)} = D^{L+j} + \cdots + D^i$$

Hence, $g(D)z(D)$ will consist of a burst of at least length L.

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Hence, $g(D)z(D)$ will consist of a burst of at least length L.

If $P<L$, $e(D)=g(D)z(D)$ is not possible!

All error bursts of length *L* and less are detectable

What about double errors?

What about *e(D)* of the type $D^j + D^i$?

We already know that if *j-i*<*L*+1, then it is detectable.

For *j-i*>*L*, more advanced theory must be used (theory of finite fields – Galois theory)

When the smoke clears, the result is

if *g(D)* is *primitive*, all double errors are detectable (if $K < 2^L - 1$)

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Some known results

With a primitive $g(D)$ times $(1+D)$, that is $g(D) = (1+D)g_p(D)$

- All single and double errors are detectable
- Burst-detecting capability of at least K
- Probability of detecting a completely random $e(D)$: 2^{-L}

Standard $g(D)$s with L=16.

- $D^{16} + D^{15} + D^2 + 1$ CRC-16 (USB)
- $D^{16} + D^{12} + D^5 + 1$ CRC-CCITT (Bluetooth, PPP)

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Some standard generator polynomials

Standard *g(D)*s with different lengths *L*.

- CRC-16 (USB) $D^{16} + D^{15} + D^2 + 1$
- CRC-CCITT (Bluetooth, PPP) $D^{16} + D^{12} + D^5 + 1$
- CRC−24A (LTE transport channels) $D^{24} + D^{23} + D^{18} + D^{17} + D^{14} + D^{11} + D^{10} + D^7 + D^6 + D^5 + D^4 + D^3 + D + 1$
- CRC-32 (Ethernet, ZIP, PNG) $D^{32} + D^{26} + D^{23} + D^{22} + D^{16} + D^{12} + D^{11} + D^{10} + D^8 + D^7 + D^5 + D^4 + D^2 + D + 1$

Luckily, there is Matlab. Play around with the CRC-class (just type ***help crc*** in Matlab)

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A note on figure and image formats

Images, JPEG

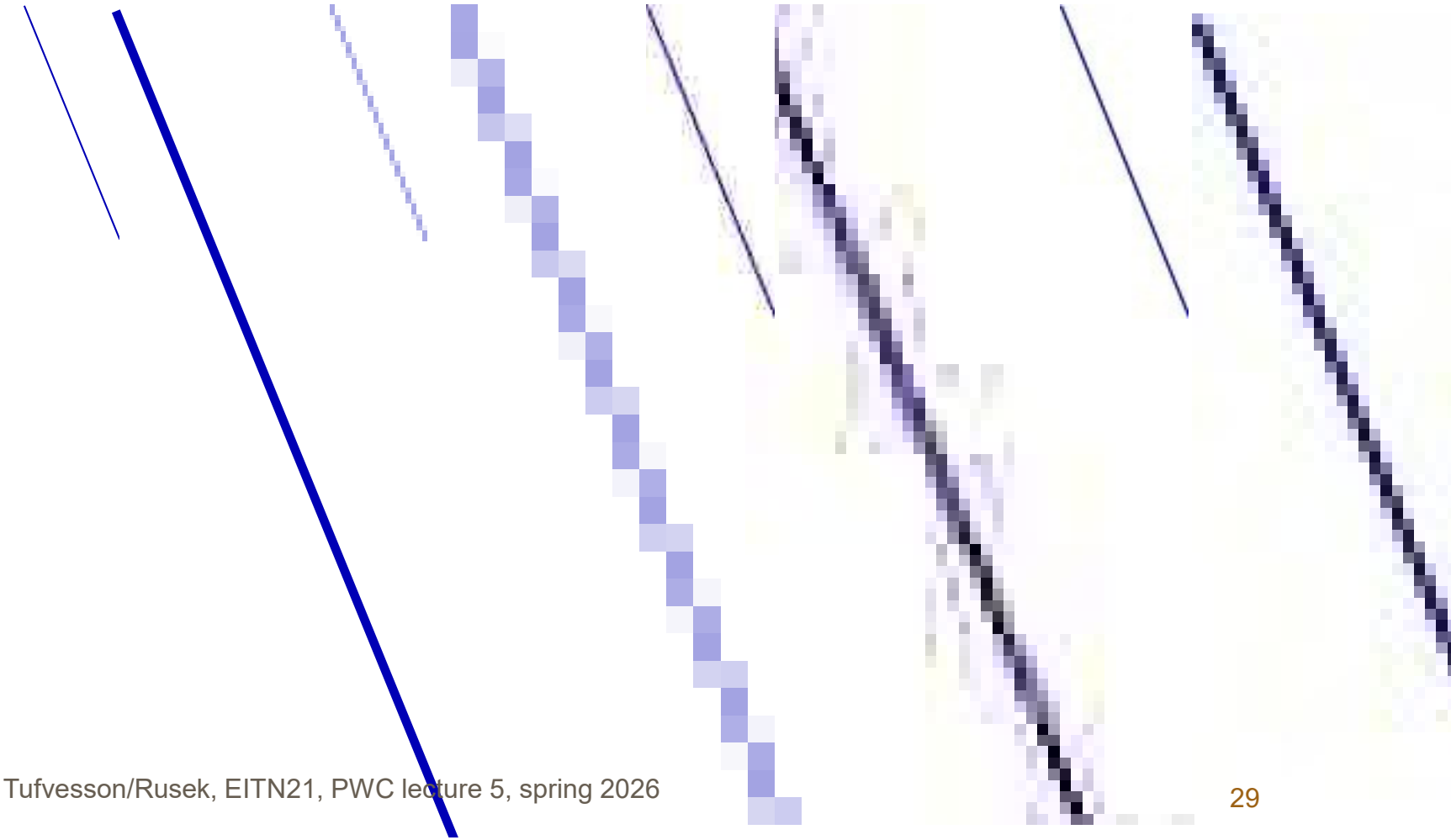
- Representing color images requires specifying the intensities Red, Blue and Green (RGB) colors.
- Digital images require huge memory for storage.
- Sophisticated image compression schemes like JPEG are employed to reduce the size of images.
- These schemes employ the properties of images and the behavior or response of human eye to reduce redundancy.
- JPEG is not a suitable to represent graphs, lines as they do not have the properties that the compression algorithms are designed for.

Other figure formats

- Vector formats (SVG, EPS)
  - Specify where lines should be drawn
  - Use this for graphs, lines (and figure text) in documents.
- Raster format (TIFF/PNG/GIF/BMP)
  - Specify each pixel value (RGB)
  - May use different levels of compression

Picture formats (original+5x magn.)

Eps vector format **png** **Bad jpeg** **Good jpeg**



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